

INFO6007 Practice Exam – Answer Guide

Based on the uploaded practice exam.

Section 1 – Multiple Choice Answers

1. b – Produces a unique outcome
2. b – Scope, Time, Cost
3. b – Iteration (Sprint)
4. a – Gantt Chart
5. c – Money actually spent to perform the work
6. c – Monte Carlo Simulation
7. b – Detailed solutions and vendor creativity are required
8. c – Transactional leadership
9. c – Operation
10. b – RACI Matrix

Question 11 – Smart City Traffic Management System

a) Critical Path and Duration

Path 1: $P \rightarrow Q \rightarrow S \rightarrow T = 3 + 5 + 6 + 2 = 16$ days

Path 2: $P \rightarrow R \rightarrow S \rightarrow T = 3 + 4 + 6 + 2 = 15$ days

Critical Path: $P \rightarrow Q \rightarrow S \rightarrow T$

Expected Project Duration: 16 days

Reason:

The critical path is the longest path in the project network. Any delay in these activities will delay the entire project.

b) Schedule and Budget Impact

If Activities Q and S take longer than expected:

- The overall project schedule may be delayed because they are on the critical path.
- Resource overuse may increase project costs and exceed the planned budget.
- Delays may also reduce productivity and create additional resource conflicts.

One practical action:

- Reallocate additional skilled resources to critical activities.

Other acceptable answers:

- Fast tracking
- Crashing
- Resource optimisation
- Frequent progress monitoring

c) Three-Level WBS Example

1. Traffic Management System

1.1 Traffic Monitoring Infrastructure

1.1.1 Smart Sensors Installation

1.1.2 CCTV Camera Setup

1.2 Software Platform Development

1.2.1 Traffic Analytics Module

1.2.2 Real-Time Dashboard

1.3 Testing and Deployment

1.3.1 Integration Testing

1.3.2 User Acceptance Testing

Question 12 – Agile E-Learning Platform

a) Managing Stakeholder Expectations in Agile

Managing stakeholder expectations is more complex in Agile because requirements can change throughout development. Stakeholders often expect quick implementation of new ideas due to Agile's flexibility and continuous feedback process.

In this scenario, the dean requests new accessibility features mid-sprint, which creates pressure on deadlines and increases team tension.

In Waterfall, scope is usually fixed earlier and changes follow a more formal process.

b) Strategies for Mid-Sprint Changes

Strategy 1:

Add the new accessibility features to the product backlog and prioritise them for a future sprint.

Strategy 2:

Conduct a meeting with stakeholders to reassess priorities, timelines, and sprint capacity.

Other acceptable answers:

- Re-estimation of sprint workload
- Splitting features into smaller increments
- Improved stakeholder communication

c) Example Project Charter Clause

“Any scope changes requested during an active sprint must be reviewed by the Product

Owner and Scrum Team. Approved changes will be prioritised in the product backlog and scheduled for future sprints unless classified as critical compliance requirements.”

Question 13 – Team Conflict and Motivation

a) Conflict Resolution Style

The most suitable conflict resolution style is Collaborating/Problem Solving.

Reason:

This approach encourages open discussion, values both senior and junior team members’ perspectives, and aims to find a mutually beneficial solution while maintaining team relationships.

b) Motivation Theory Explanation

Using Maslow’s Hierarchy of Needs:

Junior team members may feel disengaged because their esteem and belonging needs are not being met. They may feel ignored, undervalued, or excluded from important discussions, reducing motivation and confidence.

c) Leadership Practices

1. Conduct regular inclusive team discussions where all members can contribute ideas.
2. Recognise and appreciate contributions from both senior and junior team members.

Other acceptable answers:

- Mentoring and coaching
- Transparent communication
- Team-building activities
- Encouraging collaboration